From read-out geometry to in-silico stimulation: a distributed functional-connectivity signature of Alzheimer's disease

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Abstract

Resting-state functional connectivity (FC) is altered in Alzheimer's disease (AD), and the disorder is widely regarded as a distributed, network-level process; yet whether its FC signature can be reduced to a few focal sites, or is functionally irreducible to them, has not been tested causally, a question with direct consequences for targeted neuromodulation. We fit subject-specific reservoir-computing models that reconstruct each individual's lagged FC and are identifiable across subjects. Two complementary read-outs of the fitted models, the geometry of the read-out weights across patients, and the lagged FC of the model's own reconstruction, classify AD from cognitively unimpaired controls with comparable, modest accuracy. Using the same models as an in-silico testbed, we uncover a striking dissociation between the anatomy of pathology and the anatomy of effective intervention. Targeting the sites that deviate most from controls, predominantly subcortical and limbic structures (pallidum, nucleus accumbens, brainstem, amygdala), fails to revert the AD classification even at supra-physiological amplitudes, confirming that the FC signature is not reducible to those focal nodes. By contrast, selecting each patient's stimulation site by its *effect on the disease discriminant*, the site whose resonant drive shifts the reconstructed FC most toward the control distribution, achieves complete, individualised reclassification from a single site, with a real-time closed-loop controller reaching nearly the same efficacy at substantially lower dose using only causally available information. The optimal targets are predominantly cortical (prefrontal default-network, dorsal-attention and visual cortex) and highly heterogeneous across patients, indicating that effective neuromodulation requires model-informed, personalised targeting: the site to stimulate is not where the pathology concentrates but where the network is most therapeutically responsive, a prescription the fitted models make directly computable.

Introduction

Alzheimer's disease (AD) is increasingly understood as a disorder of brain networks rather than of isolated regions, in which the progressive disruption of large-scale functional interactions parallels cognitive decline [1, 2]. Resting-state functional connectivity (FC) , the inter-regional correlation structure of spontaneous BOLD activity, is among the most widely studied non-invasive readouts of this disruption, and AD is consistently associated with altered connectivity, prominently within the default-mode network [3, 4]. Yet group-level FC differences are typically modest, spatially heterogeneous, and entangled with acquisition and physiological confounds, so that FC has proven a useful but imperfect biomarker [5, 6]. Although AD is generally viewed as a distributed, network-level disorder, a more basic question remains open at the causal level: is the AD FC alteration *reducible* to a small number of regions or connections, or is it functionally irreducible to them, distributed across the network?

The answer matters well beyond diagnosis. A focal alteration would, in principle, offer a target for the emerging arsenal of network-level interventions, deep-brain, transcranial-magnetic and transcranial alternating-current stimulation, which are most effective when a small, well-defined site can be engaged [7]. A distributed alteration, by contrast, would imply that no single target suffices. This is fundamentally a causal, counterfactual question, what would happen to a patient's connectivity if a given site were corrected, that group comparisons of observational data cannot answer. Addressing it requires an individualised, generative model of each subject's dynamics that can be perturbed in silico, in the spirit of whole-brain network models and personalised brain models [8–12].

Reservoir computing offers a tractable route to such a model: a fixed random recurrent network, driven by a subject's signals, provides a rich dynamical basis from which a simple linear read-out can reconstruct that subject's activity and connectivity [13, 14]. Fitting the read-out per subject yields a compact, individualised parameterisation of the generative process, an individualised generative *surrogate* of the subject's FC, whose weights can be inspected, compared, and, crucially, modified. Here we fit such models to resting-state fMRI from AD patients and cognitively unimpaired controls, and show that the closed-loop reconstructions are faithful to and identifiable from individual connectivity, including its temporal (lagged) structure.

Building on this model, we develop the paper along a single arc, from reading out the disease to intervening on it. First, we show that two complementary read-outs of the fitted models, the geometry of the read-out weights across patients and the lagged FC of the model's own reconstruction, classify AD from controls with comparable, modest accuracy, robust to the choice of classifier and dimensionality. Second, we use the same models as an in-silico testbed and address the stimulation question in two stages. Targeting the most-deviant sites, the subcortical and limbic regions that diverge most from the control template, fails to revert the classification at any amplitude, establishing that the disease signature is not reducible to those focal nodes. Shifting the selection criterion to the *effect on the disease discriminant*, selecting, for each patient, the site whose resonant drive most efficiently shifts the FC-lag score toward the control distribution, achieves complete reclassification from a single, personalised site, with closed-loop titration reducing the perturbation cost further. The key finding is not merely that the signature is distributed, but that the locus of maximal pathology and the locus of maximal therapeutic effect dissociate: the former is subcortical and limbic; the latter is cortical and patient-specific.

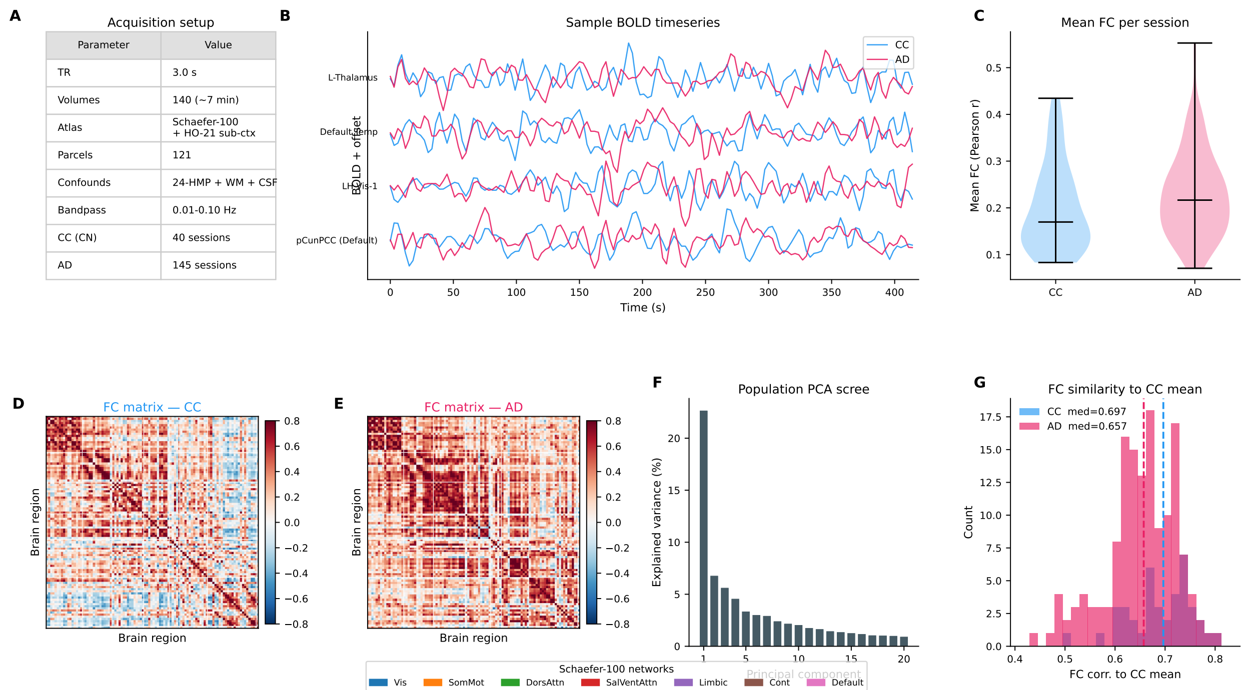

Figure 1. Dataset and resting-state functional-connectivity structure. (**A**) Acquisition and preprocessing parameters; cognitively unimpaired controls (CC) and Alzheimer's disease (AD) session counts. (**B**) Representative band-pass–filtered BOLD time series for four parcels in a CC (blue) and an AD (red) session. (**C**) Mean functional connectivity (FC; Pearson r over parcel pairs) per session, CC vs AD. The two distributions do not differ significantly (two-sided Mann–Whitney U over all 557 CC and 145 AD sessions, $p = 0.08$; the panel shows a balanced 40-session CC subsample), a weak trend toward higher mean FC in AD, consistent with the subtle, spatially diffuse group difference described below. (**D**,**E**) Group-mean FC matrices (Schaefer-100 cortical parcels, network-sorted) for CC (D) and AD (E); white lines delimit the seven Yeo networks (legend). (**F**) Population-PCA scree: variance explained by the leading components of the pooled parcel covariance. (**G**) Per-session FC similarity (Pearson r of the upper-triangular FC vector) to the CC-group mean FC, for CC and AD; dashed lines, group medians.

Results

Dataset and functional-connectivity structure

The cohort comprised resting-state fMRI sessions from cognitively unimpaired controls (CC) and patients with Alzheimer's disease (AD), parcellated into 121 cortical and subcortical regions (Fig. 1A). At the level of individual time series and their pairwise correlations the two groups were broadly similar: representative BOLD traces (Fig. 1B), group-mean FC matrices (Fig. 1D,E) and session-wise mean-FC distributions (Fig. 1C) all shared the canonical resting-state network organisation, and the population covariance was low-dimensional, a handful of principal components capturing most of the variance (Fig. 1F). Group differences were subtle and spatially diffuse rather than focal: AD sessions were on average slightly less similar to the CC-group-mean FC than CC sessions were, but the two distributions overlapped heavily (Fig. 1G), so that no single connection or region trivially separates the groups. This overlap motivated the model-based approach developed below.

A subject-specific, identifiable reservoir model of FC

For each session we fit a linear read-out from a fixed recurrent reservoir driven by the individual's signals and ran the resulting model closed-loop to generate a synthetic reconstruction (Materials and Methods). The model captured individual connectivity well: group-mean reconstructed FC matrices (Fig. 2C,D) reproduced the corresponding empirical matrices (Fig. 2A,B), and across the four group means each model FC was most similar to its own data FC, the AD model lying closest to the AD data and the CC model to the CC data (Fig. 2E). The four group-mean matrices are globally very similar (all pairwise $r > 0.93$), so this ordering is only a weak group-level effect; the decisive evidence for individual identifiability is the within- versus across-subject contrast. Crucially, reconstructions were subject-specific rather than generic: the correlation between a subject's simulated and empirical FC was far higher within- than across-subjects (Fig. 2F,G,H), establishing that the fitted models are individually identifiable. The model also reproduced the temporal structure of connectivity, the agreement between empirical and simulated lagged FC tracking the empirical decay with delay and approaching the even/odd test–retest ceiling (Fig. 2I), with reconstruction quality peaking at an intermediate read-out regularisation and exceeding a per-session baseline (Fig. 2J). Together these results establish the model as a faithful, individualised generative account of each subject's connectivity, the prerequisite for using it as an in-silico testbed.

Classification of AD from read-out geometry and reconstructed FC

We next asked whether the fitted models carry diagnostic information, comparing two read-outs of the same per-patient weights: the geometry of the read-out across patients ("G-space") and the lagged FC of the model's reconstruction ("FC-lag") (Materials and Methods). Both separated AD from CC well above chance, but with modest and comparable performance, reaching ~0.66–0.70 AUROC (Fig. 3), in the range reported for FC-based AD classifiers on comparable cohorts [15–18]. As a benchmark, an independent tangent-space (Riemannian) FC classifier with a linear support-vector machine, evaluated under *nested* 5×5 patient-level cross-validation with a train-only tangent reference, so that no test information enters the feature construction, reached AUROC 0.75 ± 0.12 (out-of-fold 0.71; Fig. 3, dotted band and ROC in Fig. 3E). The reservoir-based read-outs therefore lie in the same modest range as this leakage-free FC benchmark, at its lower end but well within its (wide) cross-validation spread, consistent with their being compact per-subject parameterisations rather than classifiers optimised for discrimination. As a further control we classified the *experimental* lagged FC directly, i.e. the lagged correlations of the empirical parcel BOLD (143 CC, 40 AD patients; same Gram-SVD embedding and leave-one-patient-out LDA/RF as the model-based FC-lag): this reached comparable accuracy (AUROC $\approx$ 0.71, peaking at $K \approx$ 20–25 and at lag 2), showing that reading out the *model's* reconstructed lagged FC neither adds nor loses diagnostic information relative to the empirical lagged connectivity, but repackages the same signal in the fitted, perturbable read-out. In a two-dimensional embedding of the FC features the CC and AD point clouds overlap substantially, yet with distinct group centroids (Fig. 3F), consistent with the modest separability that every classifier attains. Discriminability increased with the number of embedding components and plateaued near $K \approx 25$, with little benefit beyond (Fig. 3A,B), and was insensitive to the choice of classifier, a linear discriminant and

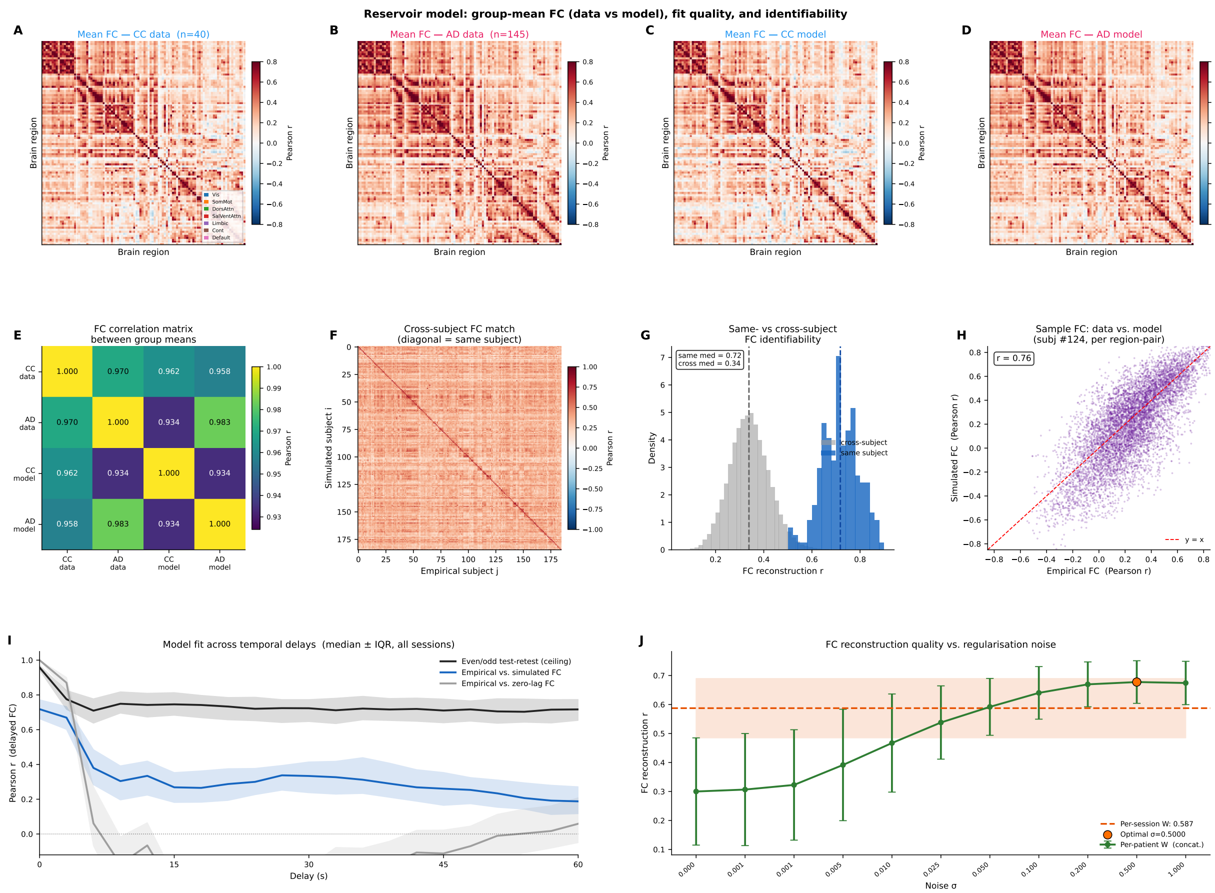

Figure 2. A subject-specific reservoir model reconstructs and is identifiable from individual functional connectivity. Per session, a fixed recurrent reservoir is teacher-forced on the PCA-projected BOLD; a linear read-out W is fit, and the model is run closed-loop (5-step drive, then free-running) to generate a simulated reconstruction $Y = W^{\top}X$. (**A**–**D**) Group-mean FC for CC and AD, from data (A,B) and from the closed-loop model (C,D). (**E**) Correlation matrix (Pearson r) between the four group-mean FCs: each group's model FC most resembles its own data FC. (**F**) Cross-subject FC matrix: correlation between simulated subject i and empirical subject j; the diagonal (same subject) is the maximum of each row. (**G**) Distribution of same-subject (diagonal) vs cross-subject (off-diagonal) FC-reconstruction r; medians dashed. (**H**) Empirical vs simulated FC for a representative subject, one point per region pair (r inset). (**I**) Model fit across temporal delays: correlation between empirical and simulated lagged FC (blue), empirical lagged vs zero-lag FC (grey), and even/odd test–retest ceiling (black); median ± IQR over sessions. (**J**) FC-reconstruction quality vs read-out regularisation noise σ for per-patient W (green); per-session baseline (orange); optimum marked.

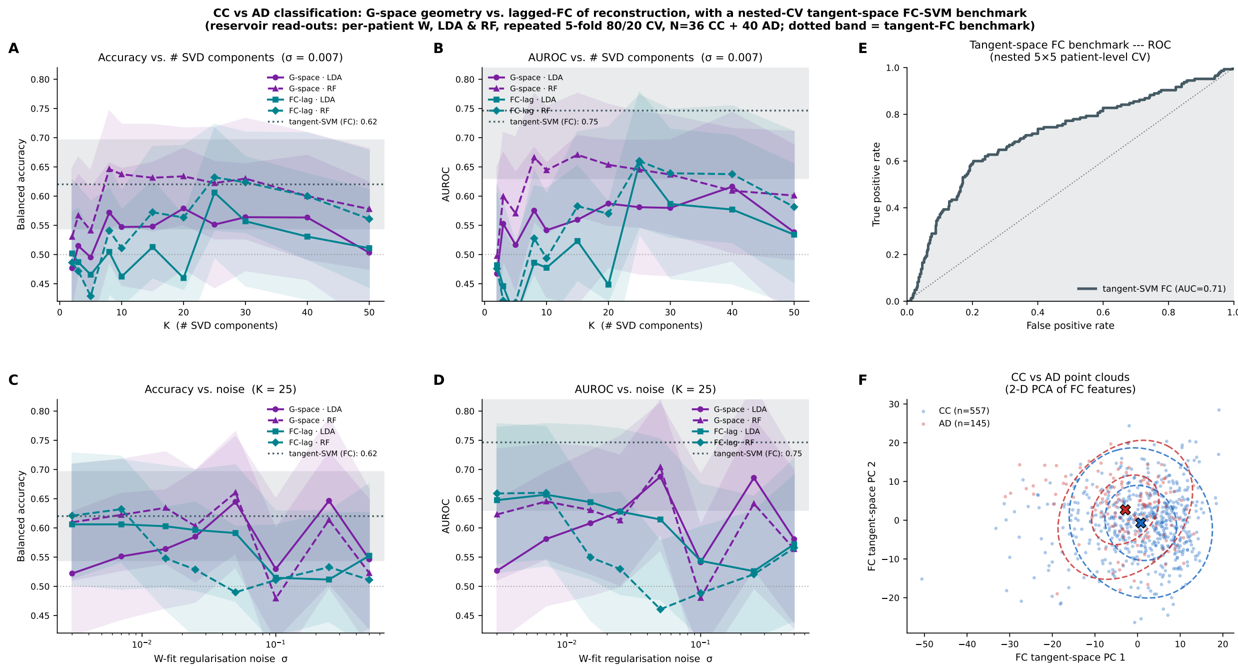

Figure 3. CC-vs-AD classification: read-out geometry vs reconstructed functional connectivity. Both classifiers derive from the *same* per-patient read-out W: "G-space" uses the SVD geometry of the projected W across patients; "FC-lag" uses the lagged-FC features (lags 0–2) of the reconstruction $Y = W^\top X$. Each is classified with a balanced Fisher LDA and a random forest (RF), evaluated by repeated (×10) stratified 5-fold (80/20) cross-validation. Lines, mean over folds; bands, ±1 SD; colour, feature space; line style/marker, classifier. (**A**,**B**) Balanced accuracy (A) and AUROC (B) vs the number of SVD components K (at the reference noise σ). (**C**,**D**) Balanced accuracy (C) and AUROC (D) vs read-out regularisation noise σ (at the reference K). Grey dotted line, chance (0.5); dark dotted line with band, an independent tangent-space (Riemannian) FC + linear-SVM classifier (AUROC 0.75 ± 0.12, mean ± SD over a *nested* 5×5 patient-level CV with a train-only tangent reference and fixed, saved fold indices for reproducibility), a strong, leakage-free FC baseline. Both reservoir-based feature spaces plateau near $K \approx 25$ and reach ∼0.70 AUROC, in the same modest range as the benchmark (at its lower end, within its wide spread), with G-space tolerating higher σ whereas FC-lag favours low σ. (**E**) Out-of-fold ROC curve of the nested-CV tangent-space FC benchmark (AUC ≈ 0.71). (**F**) Two-dimensional PCA embedding of the tangent-space FC features, coloured by group (CC blue, AD red), with 1- and 2-SD covariance ellipses and group centroids (×); the distributions overlap substantially, visualising the modest but above-chance separability that every classifier reaches.

a random forest performing similarly. The two feature spaces differed mainly in their sensitivity to the read-out regularisation: FC-lag was most accurate at low noise and degraded as the read-out was over-regularised, whereas G-space was comparatively noise-tolerant, remaining near its best over a broad range of σ (Fig. 3C,D). That two geometrically distinct read-outs reach a similar ceiling is itself consistent with a modest, distributed disease signal rather than a sharply localised one. Importantly, accuracy had not saturated at the present sample size: a learning-curve analysis over the number of patients used for fitting showed AUROC still rising with cohort size, and an empirical extrapolation projects ∼0.76 AUROC at $N = 100$ and ∼0.80 at $N = 200$ patients per group (Fig. S1), indicating that the modest accuracy reflects sample size rather than a ceiling of the representation.

In-silico stimulation: pathology site versus therapeutic target

Finally, we used the model to ask whether the AD signature can be reverted by intervention and, crucially, *which* focal target, if any, makes a single-site stimulation effective. We addressed this in two stages; as the analysis below establishes, the sites where the AD read-out deviates most from controls are not the sites where stimulation is most therapeutically effective. Throughout, "most-affected" sites are those with the largest per-site read-out deviation from the control template, i.e. the largest column norm $\|\Delta\mathbf{W}\|$ of $\Delta\mathbf{W} = \bar{\mathbf{W}}_{\mathrm{CC}} - \mathbf{W}_{\mathrm{AD}}$ (Materials and Methods). Shifting an AD read-out toward the CC template over all regions ("full-W") moved both classifiers' scores into the CC range at small interpolation strength and reclassified the large majority of AD patients as CC (Fig. 4A,C,D,F). In sharp contrast, applying the same correction to focal site sets, the single most-affected site, the five most affected, or the most-affected site together with its ten nearest neighbours, failed to revert the classification within physiological amplitude limits, shifting only a minority of patients even at much larger interpolation strengths (Fig. 4B,E; summary in Fig. 4I). Physiological diagnostics confirmed that the focal strategies were already operating at or beyond plausible amplitudes: whereas the full-W correction left signal amplitude near baseline, focal corrections required large local amplification to achieve even partial effects (Fig. 4G,H). An alternative focal protocol, injecting an oscillation at the most-affected sites, tuned to the reservoir's intrinsic frequencies, showed the same limitation, single- or few-site stimulation moving the FC-lag score only weakly and only five-site stimulation at supra-physiological amplitude approaching the decision boundary (Fig. 4J–L). Across both interpolation and oscillatory protocols, no focal intervention reproduced the effect of the distributed correction, indicating that, within the fitted models, the AD FC signature behaves as functionally distributed and is not reduced to a small set of target sites.

The anatomy of the read-out changes was nonetheless interpretable. The sites with the largest per-patient correction magnitude $\|\Delta\mathbf{W}\|$ were consistently subcortical and limbic, most frequently the bilateral pallidum and nucleus accumbens, the brainstem and the amygdala (Fig. 5A,B), each selected among the top-five sites in a large fraction of AD patients, structures of the limbic and basal-ganglia circuitry implicated in Alzheimer's disease, with the brainstem and amygdala in particular harbouring some of the earliest neurofibrillary pathology [19]. Critically, however, correcting even these maximally-deviant regions failed to revert the classification: the locus of the largest local change did not coincide with a sufficient intervention target. The disease signature is thus anatomically meaningful yet functionally distributed, not localised to the most-affected nodes.

This failure of the pathology-guided approach motivated a change of selection criterion. Rather than targeting the site of maximal read-out deviation, we asked: which site, driven at the reservoir's resonant frequency, moves each patient's FC-lag score most efficiently toward the control distribution? This *discriminant-aligned* target is a fundamentally different object from a lesion marker, it is selected by its effect on the classifier, not by its own magnitude of change, and identifying it requires the per-patient generative model. The analysis below shows that this change of criterion is decisive: the same type of focal, single-site, resonant intervention that fails at the deviation-ranked target succeeds completely at the discriminant-aligned one. Directly comparing the two site sets makes the dissociation explicit (Fig. 5C–E): the 35 distinct sites ever appearing in a patient's pathology top-5 and the 33 distinct per-patient therapy sites (personalised LDA-

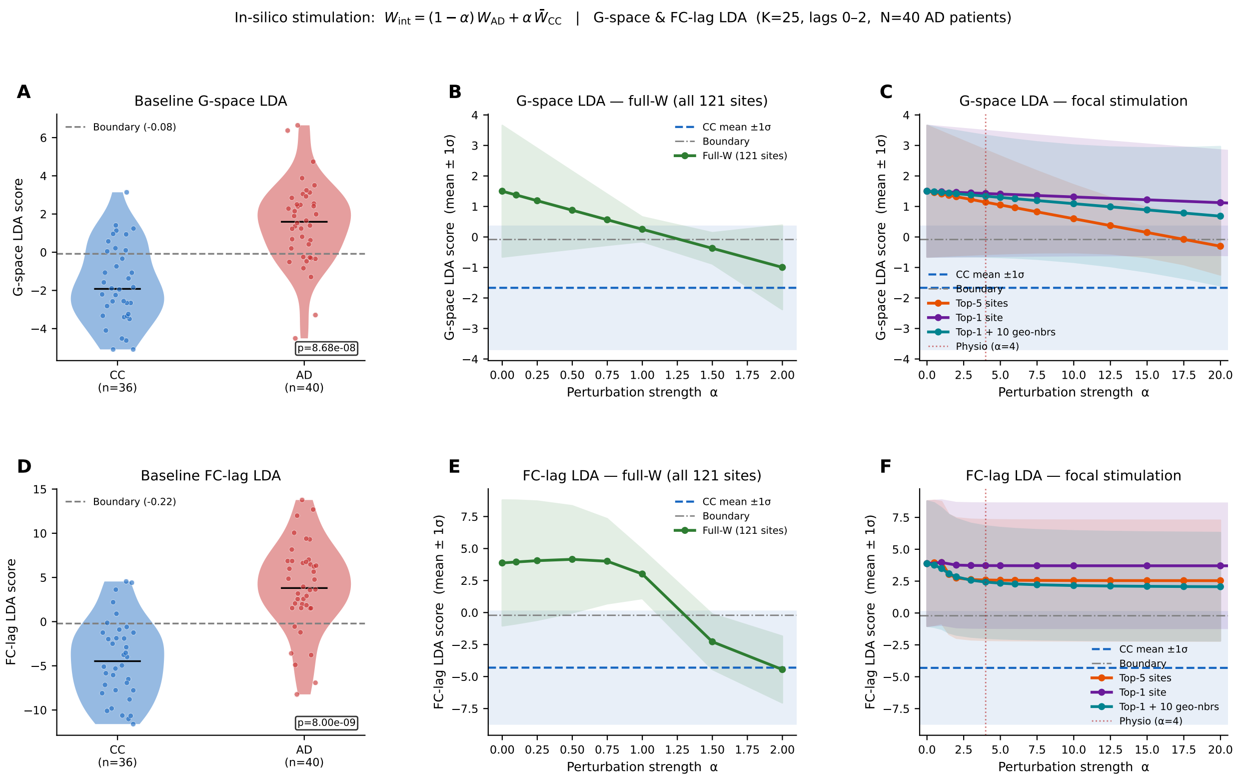

Figure 4. In-silico stimulation: targeting the most-affected sites fails; only full-W correction reverts the classification. AD read-outs are shifted toward the CC-mean read-out, $W_{\text{int}} = (1-\alpha)\,W_{\text{AD}} + \alpha\,\bar{W}_{\text{CC}}$, either over all 121 sites (full-W) or over focal site sets selected by $\|\Delta W\|$ (top-5, top-1, top-1+10 geometric neighbours). (**A**) Baseline G-space LDA scores for CC (blue) and AD (red) with the decision boundary (dashed); Mann–Whitney p inset. (**B**) G-space LDA score vs α for full-W correction (mean $\pm 1\sigma$ over AD patients); CC mean $\pm 1\sigma$ shaded; full-W enters the CC range at low α. (**C**) Same for the three focal strategies (top-5, top-1, top-1+10 geometric neighbours); none crosses the boundary within physiological amplitude. (**D**–**F**) Analogous panels for the FC-lag LDA classifier. The contrast between B/E (full-W succeeds) and C/F (focal fails at any amplitude) establishes that the AD FC signature is not reducible to the most-deviant sites.

resonant targets) identified across the 40 AD patients share only 11 sites, the great majority of each set is exclusive to its own criterion, and the therapy sites, unlike the pathology sites, have no shared anatomical hub.

Beyond identifying *where* the disease signal concentrates, the model also prescribes *how* to drive a target. Efficacy is sharply frequency-tuned: a sinusoidal drive at a focal site moves the FC-lag score toward the control distribution most efficiently when its frequency matches the reservoir's dominant eigenmode f_{eig}, the resonant drive far outperforming an off-resonance one per unit amplitude (fig. S8A,B). This frequency, $f_{\mathrm{eig}} \approx 0.077$ cycles/step, corresponds to ≈ 0.026 Hz once expressed in physical time through the TR (= 3 s) — a unit conversion, not a dependence of the resonance on TR — placing it in the infraslow band in which resting-state connectivity itself fluctuates, so the same fitted model specifies both a location (the subcortical/limbic targets of Fig. 5A,B) and an optimal driving frequency, a spatial *and* spectral prescription that descriptive analyses cannot provide. This optimum is an *interaction* of the two, however: a resonant drive reshapes the FC network-wide, yet the effect still depends strongly on the site, and the most disease-discriminative modes are not the most drivable ones, so neither frequency nor site alone suffices (fig. S7). As amplitude grows the stimulated AD score shifts steadily toward the control range and a growing fraction of patients cross the decision boundary (fig. S8C,D). A full per-patient amplitude×frequency scan confirms that reclassification concentrates in a narrow band at the resonance f_{eig}, where patients cross the boundary at the lowest amplitude and hence the smallest departure from the unstimulated simulation, though reclassification and that departure trade off, so full reversal still demands a large perturbation (fig. S9). At physiologically plausible amplitudes a focal drive does not, by itself, revert the classification (Fig. 4); its significance is methodological, demonstrating that an identifiable generative model yields concrete, testable stimulation parameters. A direct comparison of focal protocols on a common classifier (fig. S2) refines the picture. The *site-selection criterion* matters far more than the number of sites: a 2-site drive matched to the two leading reservoir eigenmodes is more efficient than single-site resonant drive, but both, selected for modal excitation or disease norm, reclassify only ~50% of patients. In contrast, choosing the target by its *effect on the discriminant* — the site whose resonant drive moves the FC-lag score most toward CC, i.e. the projection of the stimulation effect onto the LDA, a criterion we call *LDA-resonant* — is markedly more effective: a single such population-average site (the *population-average* LDA-resonant target) reclassifies ~65% of patients, and *personalising* the target (the *per-patient* LDA-resonant site, chosen from each subject's own model) reverts essentially all of them (~100%, driving the score past the CC mean; fig. S2). The most effective "drive-toward-health" sites are predominantly cortical, led by the left Default-network prefrontal cortex, with dorsal-attention and visual association cortex (fig. S4A,B); the per-patient optimal targets are highly heterogeneous, 33 distinct sites for the 40 patients, each selected by at most three (fig. S5), spanning Default-network, visual and attention cortex as well as medial-temporal and subcortical structures, so no single global target is best for all. Comparing the interventions on the closed-loop generative simulation, measured as how far each perturbs the model's own free-run dynamics relative to the unstimulated model (fig. S6), shows that reverting the classification inherently requires a large perturbation: the full exact read-out correction reverts the FC-lag classifier only with over-correction ($\alpha \gtrsim 1.4$; the exact $\alpha = 1$ replacement is insufficient because the reservoir state stays AD-driven, cf. Fig. 4). Tellingly, the personalised LDA-resonant

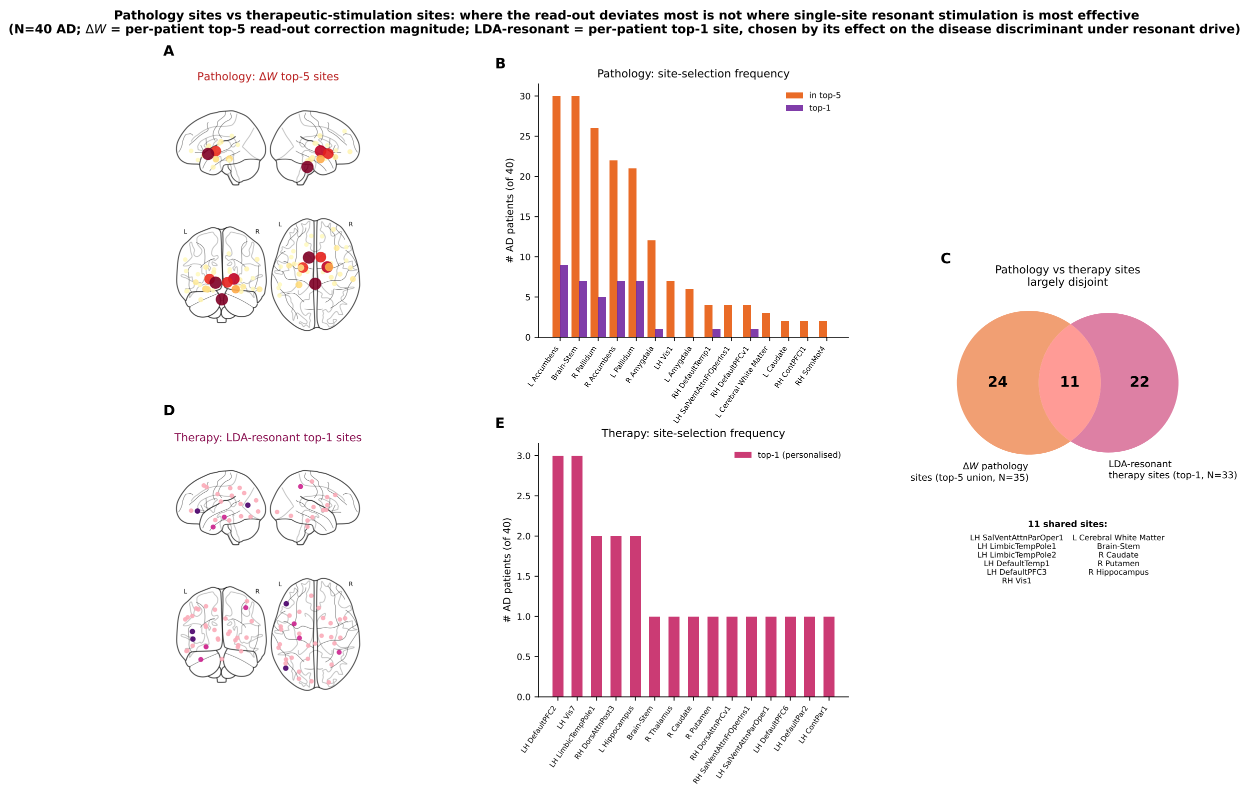

Figure 5. Pathology sites and therapeutic-stimulation sites are largely disjoint. *Pathology* sites are ranked per AD patient by the per-site read-out correction magnitude $\|\Delta\mathbf{W}\|$ (column norm of $\Delta\mathbf{W} = \bar{\mathbf{W}}_{\mathrm{CC}} - \mathbf{W}_{\mathrm{AD}}$, top-5 per patient). *Therapy* sites are each patient's personalised LDA-resonant target, the single site (top-1) whose resonant drive moves that patient's FC-lag discriminant most toward CC (Materials and Methods). (**A**) Glass-brain markers at the pathology sites (MNI centroids), coloured and sized by top-five selection frequency: deep subcortical/limbic location. (**B**) Pathology selection frequency: number of AD patients with each site among the top-five (orange) or as the single most-affected site (top-1, purple). The leading sites are subcortical and limbic, bilateral pallidum and nucleus accumbens, the brainstem and the amygdala. (**C**) Overlap between the two site sets: of the 35 distinct pathology sites (top-5 union across patients) and 33 distinct therapy sites (top-1 per patient), only 11 sites (~31% and ~33% of each set, respectively) are shared; the great majority of each set is exclusive to that criterion. (**D**) Glass-brain markers at the therapy sites, coloured and sized by top-1 selection frequency: predominantly cortical and, unlike the pathology sites, without a shared anatomical hub, each site is selected by at most three of the 40 patients. (**E**) Therapy selection frequency: the leading therapy sites are the left Default-network prefrontal cortex and visual association cortex, with no site selected by more than three patients.

and the eigenmode-coupling single-site drives perturb the simulation *near-identically* (correlation to the unstimulated model ≈ 0.32–0.34), yet only the discriminant-aligned LDA target converts that perturbation into full reclassification (100% vs ~62% for the eigenmode site): selecting the site by its effect on the discriminant, rather than by modal coupling alone, extracts more therapeutic effect per unit of dynamical perturbation, and does so by perturbing a single input site rather than rewriting all 121 read-out weights. These reversions still require supra-physiological amplitudes and the personalised target is selected on each patient's own model (an upper bound on achievable personalisation); their significance is to show that an identifiable, *linear* per-subject model converts "where and how to stimulate to move a patient toward health" into a directly computable, individualised prescription.

To isolate what makes a *single-site* intervention work, we contrasted the two ways of acting on one node at the same most-affected target (Fig. 6): the *theoretical* read-out correction, which interpolates that node's read-out column toward the control mean ($\mathbf{W}_{\text{int}} = (1 - \alpha)\mathbf{W}_p + \alpha\bar{\mathbf{W}}_{\text{CC}}$), and the *physical* resonant drive ($A \sin(2\pi f_{\text{eig}} t)$). The theoretical single-column correction is essentially inert: even pushed to $\alpha = 50$ it never reverts the classification (it stays at the ~18% baseline), while progressively distorting the simulation (distance from the unstimulated FC rising to ~0.6), a single read-out weight has negligible leverage on the spatially distributed FC-lag signature. The physical resonant drive instead reclassifies patients by engaging the network-wide resonant mode, but *where* it is applied is decisive: at the ΔW most-affected (subcortical) site it reaches only ~50–57% even at high amplitude, whereas at each patient's discriminant-aligned (LDA-resonant) site it reverts *all* of them (Fig. 6A). Either way its distance from the original FC jumps to ~0.7 at the smallest effective amplitude and saturates (Fig. 6B,C). At one site, then, a read-out edit is inert and only a resonant drive at the right site cures, at a fixed, large perturbation cost that the closed-loop control below is designed to minimise.

Because the fitted model is a closed-loop dynamical system, the same FC-lag score that diagnoses a patient can serve as the feedback signal for a *real-time, causally valid* adaptive intervention: rather than an offline oracle with access to the full future trajectory, we titrate the resonant drive's amplitude online, every few steps, from a sliding window of the model's own past output (Materials and Methods). At the fixed, personalised LDA-resonant site, this online controller reclassifies 39/40 (97.5%) of AD patients, nearly matching the 100% reached by a fixed open-loop drive, while cutting the mean stimulation amplitude by ~44% ($A = 6 \to 3.3$) and the departure from each subject's own dynamics by ~7% among patients reclassified under both protocols (distance $0.69 \to 0.64$; Fig. 6D–F). Online amplitude titration therefore converts the costly fixed drive into a substantially cheaper, individualised protocol that reaches nearly the same in-silico reversal closer to the patient's own resting dynamics, using only information available causally, in real time.

Discussion

We fit identifiable, subject-specific reservoir-computing models of resting-state functional connectivity and used them both to read out and to intervene on the Alzheimer's signature. Two complementary read-outs of the fitted models, the geometry of the read-out weights across patients and the lagged FC of the model's

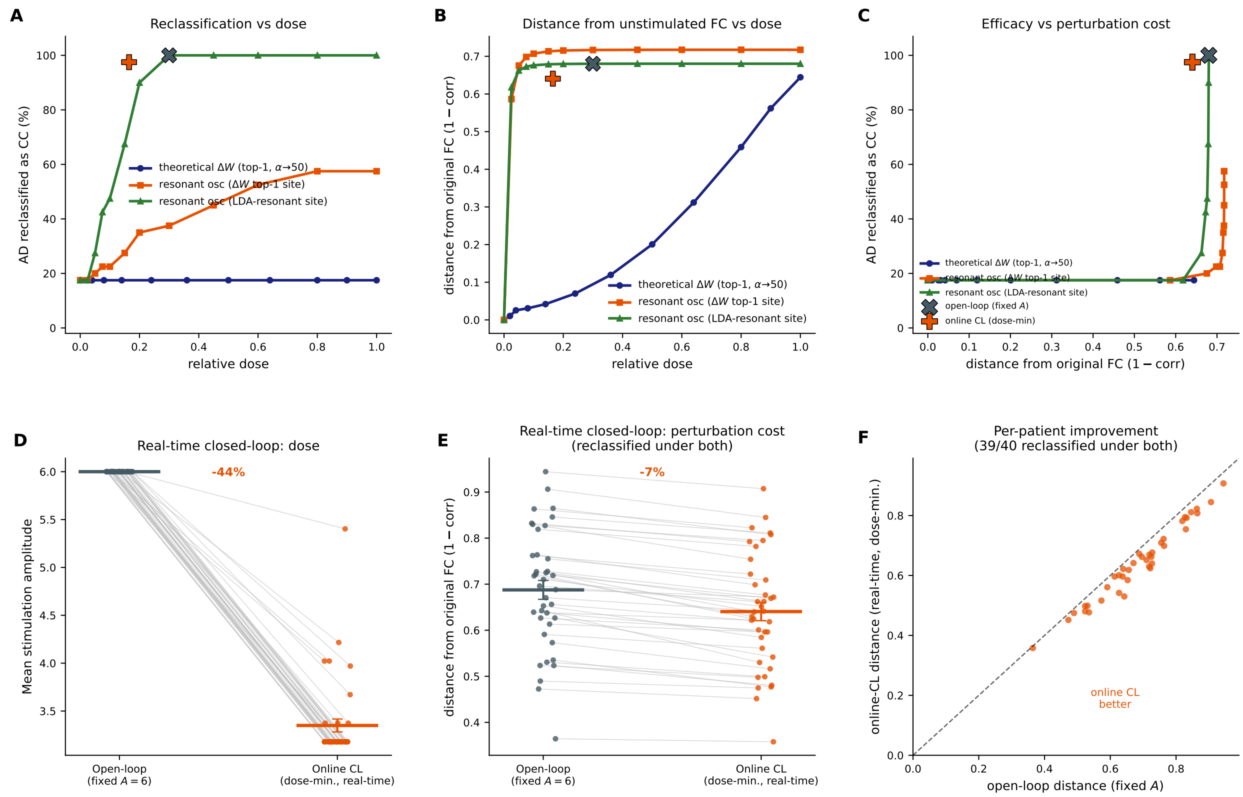

Figure 6. Single-site stimulation and closed-loop control ($N = 40$ AD; distance = 1−corr between the stimulated and unstimulated closed-loop free-run FC). *Top row, single-site dose-response* comparing a *theoretical* read-out correction (ΔW interpolation of the top-1 most-affected column toward the CC mean, $\alpha \in [0, 50]$), a *resonant* drive $A \sin(2\pi f_{\mathrm{eig}} t)$ at that same ΔW site, and a resonant drive at each patient's *LDA-resonant* (discriminant-aligned) site ($A \in [0, 20]$). (**A**) Reclassification vs normalised dose: the single-column ΔW correction is inert (flat at the 18% baseline even at $\alpha = 50$); the resonant drive at the ΔW site saturates near ~57%; the resonant drive at the LDA-resonant site reverts *all* patients (100%). (**B**) Distance from the original FC: the ΔW correction distorts only gradually, whereas both resonant drives jump to ~0.7 at the smallest effective amplitude and saturate (the oscillation dominates the network-wide mode). (**C**) Efficacy vs perturbation cost: only the LDA-resonant drive converts perturbation into full reclassification. The closed-loop *operating points* (patient means; grey × open-loop, orange + online closed-loop) are overlaid on **A**–**C**: the online closed-loop point sits at a substantially smaller dose than the open-loop point, for a comparable, though not quite complete (97.5% vs 100%), reclassification rate and a modestly smaller distance. *Bottom row, real-time closed-loop control at the LDA-resonant target* (the same curing site as the green curve of **A**), comparing a fixed *open-loop* drive ($A = 6$) to an *online closed-loop* controller that titrates amplitude every few steps from a sliding-window estimate of the FC-lag biomarker computed only from the model's own past output, no oracle access to the future trajectory (Materials and Methods). (**D**) Per-patient stimulation amplitude (dose): online control lowers the mean dose from 6.0 to 3.3 (~44% lower) while nearly matching efficacy (39/40, 97.5%, vs 40/40 open-loop). (**E**) Distance from baseline among patients reclassified under both protocols ($n = 39$): online control lowers the mean distance by ~7% ($0.69 \rightarrow 0.64$). (**F**) Per-patient: most subjects lie below the diagonal, i.e. real-time control reaches nearly the same reversal at a smaller perturbation cost, the in-silico analogue of adaptive (closed-loop) neuromodulation, using only information available causally, online.

own reconstruction, classified AD from controls with comparable, modest accuracy. The central finding of the stimulation analysis is a dissociation between the anatomy of pathology and the anatomy of effective intervention. Targeting the sites of greatest read-out deviation from controls, predominantly subcortical and limbic, fails to revert the classification even at supra-physiological amplitudes, confirming that the FC signature is not concentrated in those regions. Selecting the target instead by its effect on the disease discriminant under a resonant drive (the LDA-resonant criterion) achieves complete reclassification from a single, individualised site, identifying for each patient the node where stimulation is most therapeutically effective. The two site sets are anatomically distinct: pathology concentrates in subcortical and limbic hubs, whereas therapeutic leverage resides in cortical nodes, predominantly prefrontal default-network and dorsal-attention cortex, that propagate network-wide corrections through their connectivity. Whether this dissociation reflects the biology of AD or the representational geometry of the model is a question our framework raises but cannot, on its own, settle.

This conclusion is consistent with the view of AD as a network, or "disconnection", disorder rather than a regionally circumscribed one [1, 2], and extends it with a causal, counterfactual argument that observational group comparisons cannot make: within an individualised generative model, targeting where the pathology is greatest is not equivalent to targeting where stimulation is most effective. The practical implication for neuromodulation is twofold. First, a single-target intervention *can* revert a distributed connectivity deficit, but only when the target is selected for its effect on the disease discriminant rather than for its pathological magnitude; this distinction is not accessible without a per-patient generative model. Second, the effective site is highly patient-specific (33 distinct parcels across the 40 AD patients), so no single anatomical landmark serves all individuals: personalised, model-informed targeting is required. Both the per-patient discriminant-aligned sites and the population-average LDA-resonant target (left prefrontal default-network cortex) are consistent with loci explored in non-invasive AD neuromodulation [7, 20], lending translational plausibility to the model-derived prescription. For patients whose optimal site is inaccessible or whose response at a single site is insufficient, the present framework also provides a way to design distributed, multi-site or network-shaped stimulation strategies in silico before any intervention, complementing model-based approaches that predict stimulation spread in whole-brain models [21–24].

The anatomy of the read-out deviations grounds these conclusions biologically. The regions whose read-out departed most from the control template were predominantly subcortical and limbic, bilateral pallidum and nucleus accumbens, brainstem and amygdala, structures of the limbic and basal-ganglia circuitry, several of which (notably the brainstem and amygdala) harbour some of the earliest neurofibrillary pathology in Braak staging [19]. These maximally-deviant regions were not, however, viable focal intervention targets: the locus of greatest pathological change and the locus of greatest therapeutic leverage dissociate cleanly. A plausible mechanistic account is that pathology originating in limbic hubs propagates along structural and functional connections, distributing its imprint on connectivity across the network [25, 26]; the discriminative signal then resides in the network-wide pattern, so that correcting the seed regions alone is insufficient. The cortical nodes that do provide therapeutic leverage, prefrontal default-network, dorsal-attention and visual cortex, are precisely those best positioned to redistribute a corrective perturbation network-wide through their dense long-range connectivity, and they interface with the subcortical structures of greatest pathological deviation via established DMN–limbic pathways. This dissociation between lesion locus and effective intervention site is consistent with the network-medicine principle illustrated most clearly for deep-brain stimulation [7]: the optimal target is not the damaged node but the network node most effectively coupled to the symptom circuit. The per-patient heterogeneity of the optimal sites (33 distinct parcels for 40 patients, Fig. S5) further argues that no universal anatomical landmark suffices, a result consistent with the clinical heterogeneity of AD and with the emerging literature on connectivity-guided personalised neuromodulation.

A distinctive feature of the framework is that it prescribes not only where but how to stimulate. The driving frequency that most efficiently moved a target toward the control distribution coincided with the model's intrinsic resonance (fig. S8), connecting our results to the broader principle of frequency-specific neuromodulation. Stimulation efficacy is well known to depend on frequency, and entraining endogenous rhythms is an active therapeutic strategy in AD, most prominently 40 Hz "gamma" sensory and electrical stimulation, which reduces pathology and improves function in mouse models and early human studies [27, 28]. Our prescription operates at a far slower, infraslow timescale, the band in which resting-state connectivity fluctuates, rather than in the gamma range, reflecting the BOLD-derived nature of the model; we therefore make no claim about a specific deliverable protocol. The conceptual contribution is that an identifiable, subject-specific model turns "at what frequency should one stimulate?" into a computable quantity tied to the network's own dynamics, complementing connectome-model approaches that predict the spatial spread of stimulation [21, 23]. A practical advantage is that, because the reservoir dynamics and the read-out are *linear*, the response to a periodic drive, and hence the optimal stimulation site and frequency, follows directly from the system's transfer function and eigenspectrum, rather than from an exhaustive search: the resonance we observe is precisely the peak of this linear frequency response. The model thus affords a closed-form, analytically tractable prescription of optimal stimulation parameters. Notably, this optimum is an *interaction* of site and resonance that no single-factor heuristic captures: a resonant drive reshapes the FC network-wide, yet the reclassification still depends strongly on the site (fig. S7A), the disease-discriminative modes are not the drivable ones (the dominant, most-drivable mode is only weakly discriminative; fig. S7B), and steering by a discriminative-but-undrivable frequency, alone or combined with the resonance, does not help (fig. S7C,D). Selecting the target by its *effect on the discriminant under a resonant drive* (the LDA-resonant criterion) is what jointly resolves drivability and direction, and is why it outperforms modal-coupling or disease-deviation selection. Reassuringly, when the target is selected for its effect on the disease discriminant, the leading "drive-toward-health" site falls in the Default-network prefrontal cortex (fig. S4), a hub of the network most consistently disrupted in AD [3, 4] and a prime locus for non-invasive neuromodulation, e.g. precuneus/Default-network stimulation trials [20], lending biological plausibility to the model-derived prescription. The heterogeneity of the per-patient optimal targets further argues that effective targeting must be individualised rather than one-size-fits-all, consistent with connectivity-guided personalised stimulation [7]. Because the fitted model is a dynamical system, the same biomarker that scores a patient can also *close the loop* in real time: titrating the drive's amplitude online against a sliding-window FC-lag estimate, using only the model's own past output rather than an offline oracle, reclassifies 97.5% of patients (nearly matching the 100% open-loop rate) while cutting both the stimulation dose (∼44%) and the departure from their own dynamics (∼7% among patients reclassified under both protocols) relative to a fixed open-loop drive (Fig. 6), the in-silico counterpart of adaptive (closed-loop) neuromodulation, in which a causally available, minimal-dose perturbation that crosses the diagnostic boundary, rather than a supra-physiological open-loop drive, is sought. We emphasise, however, that these are *model-level* prescriptions: the reservoir is fit to infraslow BOLD connectivity, not electrophysiology, so an "intervention" here is an abstract input to the regional FC dynamics, and its optimal spatial pattern and (infraslow) frequency should be read as targets at the level of regional drive, not as cellular or oscillatory mechanisms. With that caveat, the focal, slowly-modulated character maps in spirit onto closed-loop, biomarker-guided neuromodulation; translating it to any specific technology, and validating it against electrophysiological or longitudinal outcomes, is left to future work.

More broadly, the study illustrates a general methodology: an identifiable individualised generative surrogate of a subject's dynamics, fit once, can serve simultaneously as a biomarker substrate and as a counterfactual testbed. Because the read-out is a compact per-subject parameterisation, the same fitted object that classifies the disease can be perturbed to ask what would reverse it, a loop from decoding to intervention that is difficult to close with descriptive analyses alone, and that transfers directly to other conditions and to stimulation planning.

Several limitations qualify these conclusions. The cohort is cross-sectional and of modest size, and the classifiers, while honestly cross-validated, reach only moderate accuracy and embed each subject relative to the group (a transductive embedding); per-subject read-outs were estimated from limited data, adding fitting noise that we characterised but did not eliminate. The reservoir is a phenomenological rather than biophysical model, with fixed random recurrent weights, and our "stimulation" is an idealised modification of the read-out or its input rather than a literal physical protocol; the mapping between in-silico amplitude and a deliverable dose is therefore approximate, and the anatomical neighbourhoods rely on coarse parcel centroids. These factors bear on quantitative details but not on the central, qualitative contrast between the distributed and focal interventions, which is large and consistent across classifiers and protocols.

Future work should test the distributed signature longitudinally and in larger, multi-site cohorts, replace the phenomenological reservoir with more biophysically grounded generative models, and connect the in-silico targets to realisable stimulation montages so that distributed and multifocal strategies can be evaluated explicitly. Formally testing whether the distributed FC signature follows the predicted spread of pathology along network connections, rather than merely co-localising with the most-affected hubs, would turn the reconciliation proposed above into a falsifiable hypothesis.

Materials and Methods

Data and preprocessing. Resting-state functional MRI sessions were downloaded from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu), comprising cognitively unimpaired controls (CC) and patients with Alzheimer's disease (AD). Each session comprised 140 volumes (repetition time TR = 3.0 s, ~7 min). Images were minimally preprocessed and parcellated into $N_{\mathrm{p}} = 121$ regions of interest, the 100 cortical parcels of the Schaefer 7-network atlas [29] plus 21 Harvard–Oxford subcortical labels. All 121 parcels (cortical and subcortical) enter the reservoir fit, the classifiers and the stimulation analyses throughout; the network-sorted group-mean FC matrices of Fig. 1D,E display the 100 cortical parcels only, for which the Yeo 7-network assignment is defined. Region-wise BOLD time series were denoised by regression of 24 head-motion parameters and mean white-matter and cerebrospinal-fluid signals, and band-pass filtered to 0.01–0.10 Hz. To balance the two groups during reservoir fitting, classification and stimulation we sampled 40 CC sessions to match the 40 AD subjects passing quality control; grouping by subject (one session each) gives 36 CC and 40 AD subjects. The full pool (145 CC and 41 AD subjects; 561 and 151 sessions) was used for the cohort-size scaling and for the tangent-space SVM benchmark. The latter used the globally signal-regressed sessions (557 CC + 145 AD sessions passing a >100-volume filter) under *nested* 5×5 subject-grouped cross-validation: Ledoit–Wolf covariances projected to the tangent space at a *train-only* reference mean, then standardised, PCA-reduced, and classified by a linear SVM or logistic regression, with the embedding dimension, penalty and classifier chosen in the inner loop and evaluated on held-out subjects in the outer loop. Sessions were grouped by subject, and per-subject quantities used the first available session unless otherwise stated; Table S1 summarises the cohort used in every analysis.

Population principal-component projection. All region time series were pooled and mean-centred, and the leading $N_{\mathrm{PC}} = 50$ eigenvectors $\mathbf{V}_{50}$ of the $N_{\mathrm{p}} \times N_{\mathrm{p}}$ covariance were retained. For each session with signal $\mathbf{s} \in \mathbb{R}^{N_{\mathrm{p}} \times T}$ we defined the PCA-reconstructed *target* $\hat{\mathbf{s}} = \mathbf{V}_{50}\mathbf{V}_{50}^{\top}\mathbf{s}$, which removes idiosyncratic high-order variance while preserving the dominant spatiotemporal structure. Empirical functional connectivity (FC) was the Pearson correlation matrix of $\hat{\mathbf{s}}$. The $N_{\mathrm{PC}} = 50$ projection defines only the reservoir's training *target*; the read-out and all FC quantities remain in the full $N_{\mathrm{p}} = 121$ region space, so the regional dimensionality is never reduced below 121.

Reservoir model. We used a fixed random recurrent network of $N = 2000$ rate units with state $\mathbf{x}(t)$, internal weights $\mathbf{J}$ (Gaussian, rescaled to spectral radius 0.95), and fixed random input weights $\mathbf{J}_{\mathrm{in}}$ ($\sigma_{\mathrm{in}} = 0.01$; the subscript "in" distinguishes these untrained input weights from the trained read-out $\mathbf{W}$). The network is a *linear*, deterministic reservoir: the update below is linear in $\mathbf{x}$ and no endogenous noise is injected into the dynamics (regularisation noise enters only the read-out fit, Eq. 2). Units evolved in discrete time ($\mathrm{d}t = 0.005$) as

$$\mathbf{x}(t + \mathrm{d}t) = e^{-\mathrm{d}t/\tau}\,\mathbf{x}(t) + \left(1 - e^{-\mathrm{d}t/\tau}\right)\left(\mathbf{J}\,\mathbf{x}(t) + \mathbf{J}_{\mathrm{in}}\,\mathbf{u}(t)\right), \tag{1}$$

with a short membrane time constant τ (effectively instantaneous update), and a linear read-out $\mathbf{y}(t) = \mathbf{W}^{\top}\mathbf{x}(t)$ in the $N_{\mathrm{p}} = 121$-dimensional region space. The same reservoir (identical $\mathbf{J}, \mathbf{J}_{\mathrm{in}}$) was reused for every session.

Read-out fitting. For each session the reservoir was teacher-forced by feeding the target as input, $\mathbf{u}(t) = f\,\hat{\mathbf{s}}(t)$ with feedback gain $f = 0.1$, and the hidden states $\mathbf{X} \in \mathbb{R}^{T_{\mathrm{eff}} \times N}$ were collected after discarding the first 10 steps. The read-out $\mathbf{W} \in \mathbb{R}^{N \times N_{\mathrm{p}}}$ was obtained by ridge-like regularised least squares with additive Gaussian noise of scale σ in the regressor,

$$\mathbf{W} = \left(\mathbf{X} + \mathbf{E}\right)^{+}\mathbf{Y}, \qquad E_{ij} \sim \mathcal{N}(0, \sigma^2), \tag{2}$$

where $\mathbf{Y}$ is the time-aligned target and $(\cdot)^+$ the Moore–Penrose pseudoinverse. The noise scale σ acts as the ridge-regularisation parameter of the read-out fit (larger σ, stronger regularisation); $\sigma = 0.025$ unless swept (Fig. 3).

Closed-loop reconstruction. The model has two operating modes. In *teacher-forced* (open-loop) mode the real target drives the input, $\mathbf{u}(t) = f\,\hat{\mathbf{s}}(t)$; this mode is used to fit $\mathbf{W}$ and to compute the classifier features. In *closed-loop* (free-run) mode the read-out feeds back its own output, $\mathbf{u}(t) = f\,\mathbf{y}(t)$, generating an autonomous reconstruction. To validate the generative model (Fig. 2) the reservoir was reset and run closed-loop: the real target drove the network for the first 5 steps, after which it free-ran for the remainder of the session. The simulated reconstruction $\mathbf{Y}_{\mathrm{sim}}$ and its (lagged) FC were compared to the empirical signals. Subject identifiability was quantified by the matrix of correlations between simulated subject i and empirical subject j, comparing the same-subject (diagonal) and cross-subject (off-diagonal) distributions. Both intervention families are evaluated consistently in both modes: a read-out interpolation ($\mathbf{W}_{\mathrm{int}}$) and an input oscillation (A sin) each alter the same reservoir, and each is scored by the classifiers on the teacher-forced FC-lag features *and* assessed for its departure from baseline on the closed-loop free-run FC (Fig. 6, figs. S6, S9); the two are therefore compared on a common footing, not one in open- and the other in closed-loop.

Read-out "G-space". Each read-out was projected onto the row space of its own reservoir states: with $\mathbf{V}_k$ the top $K_{\mathrm{PC}} = 200$ right singular vectors of $\mathbf{X}$, the projected read-out is $\widetilde{\mathbf{W}} = \mathbf{W}^\top \mathbf{V}_k \mathbf{V}_k^\top$, vectorised to $\mathbf{w}_i$. Stacking patients, centring, and taking the SVD of the centred matrix yielded patient coordinates ("G-scores") in a low-dimensional read-out-archetype space; the first K components were used for classification. The projection rank $K_{\mathrm{PC}} = 200$ and the low-dimensional sufficiency of the archetype basis are justified by a training/test reconstruction-error analysis over (K_{PC}, M) (fig. S3).

Lagged-FC features. From the reconstructed signals $\mathbf{Y} = \mathbf{W}^\top \mathbf{X}$ we computed lagged correlation matrices for lags 0–2 (zero-lag upper triangle plus full lag-1 and lag-2 matrices), concatenated into a feature vector per patient. A Gram-matrix SVD across patients (the transductive embedding) gave the "FC-lag" coordinates, of which the first K were used.

Classification. CC vs AD was classified with two algorithms on each embedding: a balanced Fisher linear discriminant (LDA; majority class sub-sampled to the minority within each training set) and a random forest (RF; 300 trees, $\sqrt{\cdot}$ features, minimum leaf size 3, balanced class weights). Performance was estimated by repeated stratified 5-fold cross-validation (80/20 train/test, 10 repeats), reporting balanced accuracy and the area under the ROC curve (AUROC) as the mean ± standard deviation over folds. We swept the embedding dimension $K \in [2, 50]$ and the read-out noise $\sigma \in [0.003, 0.5]$; reference slices were chosen by the marginal-mean AUROC across classifiers and the complementary axis. The SVD embeddings were built transductively (across all patients) while the classifiers were trained and tested on disjoint folds.

In-silico stimulation. The model was used as a counterfactual testbed by shifting each AD read-out toward the CC-mean read-out, $\mathbf{W}_{\mathrm{int}} = (1-\alpha)\,\mathbf{W}_{\mathrm{AD}} + \alpha\,\bar{\mathbf{W}}_{\mathrm{CC}}$, applied either to all 121 columns (sites; full-W) or to focal subsets. Here α is the read-out interpolation strength: $\alpha = 0$ leaves the unmodified AD read-out, $\alpha = 1$ replaces it entirely by the CC-mean read-out $\bar{\mathbf{W}}_{\mathrm{CC}}$, and $\alpha > 1$ is over-correction. We nonetheless explore $\alpha > 1$ because the exact $\alpha = 1$ replacement can be insufficient: the reconstruction is closed-loop, so the reservoir state driven by the AD input keeps the FC partly AD-like even under the CC read-out, and crossing the decision boundary can require pushing past $\bar{\mathbf{W}}_{\mathrm{CC}}$ (cf. fig. S6). (α is distinct from the oscillation amplitude A below.) Focal sites were ranked per patient by the column norm of $\Delta\mathbf{W} = \bar{\mathbf{W}}_{\mathrm{CC}} - \mathbf{W}_{\mathrm{AD}}$: the top-5, the top-1, and the top-1 site plus its 10 nearest neighbours in MNI space (parcel centroids from the atlas). For each α the perturbed read-out was re-scored by both classifiers, and the AD→CC reclassification rate was the fraction crossing the CC/AD decision boundary. We report a heuristic reference amplitude of $\alpha = 4$ for summary comparisons; it is a convenient common operating point, not a biophysically calibrated limit, and the full dose-response is shown throughout.

Oscillatory stimulation. As an alternative focal protocol, a sinusoid $A \sin(2\pi f t)$ (amplitude A in input units, in the same units as $\mathbf{J}_{\mathrm{in}}\mathbf{u}$ and distinct from the read-out interpolation strength α above) was injected as additional input at the top-k sites ($k = 1, 2, 5$) during the teacher-forced pass, with $\mathbf{W}$ unchanged; the oscillation propagates through the recurrent dynamics and reshapes the reconstructed FC, which was re-scored by the FC-lag classifier. Two candidate drive frequencies were compared: the leading complex eigenmode of $\mathbf{J}$ (model-intrinsic, requiring the eigendecomposition of $\mathbf{J}$) and the peak of the empirical power spectrum of the AD reservoir states (data-driven, obtained by FFT without inspecting $\mathbf{J}$); the two agree closely. Because $\mathbf{J}$ (and hence its eigenspectrum) is fixed and shared across subjects, the resonance f_{eig} is a property of the model, not a subject-specific fit. Frequencies were swept over $f \in [0, 0.5]$ cycles/step and amplitude up to *supra-physiological* values — i.e. beyond the RMS of the endogenous teacher-forcing drive (the target scaled by the feedback gain 0.1), so that the stimulated-to-baseline amplitude ratio $\mathrm{RMS}_{\mathrm{pert}}/\mathrm{RMS}_{\mathrm{base}} > 1$ (Fig. 4G). The state-level response of the linear reservoir to a periodic drive is governed by the transfer function $(\mathrm{i}\omega\mathbf{I} - \mathbf{J})^{-1}\mathbf{J}_{\mathrm{in}}\mathbf{b}$, which is large only when the drive frequency is near an eigenmode *and* the spatial input pattern $\mathbf{b}$ overlaps that eigenmode; a single site thus resonates cleanly at f_{eig}, whereas the summed pattern of several sites overlaps other modes and de-tunes the state-level resonance. Because the FC-lag score is a nonlinear functional of the read-out of the states rather than the excitation amplitude itself, this state resonance does not translate into a correspondingly sharp multi-site frequency optimum: we verified that selecting the five sites of maximal modal excitation $|\mathbf{w}^{\mathsf{H}}\mathbf{J}_{\mathrm{in}}|$ (with $\mathbf{w}$ the leading left eigenvector of $\mathbf{J}$) does *not* recover an f_{eig} optimum. The clean spectral prescription is therefore specific to single-site drive.

Physiological diagnostics. For every strategy and α we recorded the signal-amplitude ratio at stimulated sites, $\mathrm{RMS}_{\mathrm{pert}}/\mathrm{RMS}_{\mathrm{base}}$, and the mean absolute FC at those sites, to verify that reclassification (where it occurred) was achieved within plausible amplitude bounds.

Statistics and software. Group comparisons used two-sided Mann–Whitney U tests; distributions are shown as violins with medians or as mean ± SD/IQR as stated. Analyses were implemented in Python (NumPy, scikit-learn, nilearn, Matplotlib).

Ethics. This work is a secondary analysis of de-identified, publicly available human imaging data obtained from ADNI. The ADNI study was approved by the institutional review boards of all participating sites, and all participants (or their authorised representatives) provided written informed consent at enrolment; no new data were acquired for the present study.

Data and code availability. The imaging data analysed here were downloaded from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). ADNI data are available to qualified researchers upon registration and acceptance of the ADNI Data Use Agreement; in accordance with that agreement, the raw and derived imaging data are not redistributed here. The exact cohorts and sample sizes used in every analysis are listed in Table S1. All analysis and figure-generation code is openly available at github.com/cristianocapone/AD-reservoir-FC-stimulation.

Acknowledgments. Data used in preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf. Data collection and sharing for ADNI was funded by the ADNI (National Institutes of Health grant U01 AG024904) and DOD ADNI (Department of Defense award number W81XWH-12-2-0012).

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Supplementary Materials

Table S1. Cohorts used in each analysis. Minimally preprocessed resting-state fMRI (121 parcels, 140 volumes, TR = 3 s). "Subjects" are unique participants; "sessions" are individual scans (a subject may contribute several). Per-subject reservoir quantities use the first valid session.

Analysis	Controls (CC)	AD	Cross-validation / unit
Available pool	145 subj / 561 ses	41 subj / 151 ses	,
Data overview, FC similarity (Fig. 1)	561 sessions	151 sessions	per session
Reservoir fit & FC validation (Fig. 2); G-space & FC-lag classifiers (Fig. 3A–D); all in-silico stimulation (Figs 4–7, S2–S10)	36 subjects	40 subjects	one session/subject; 40 CC sessions sampled to balance 40 AD (36 unique CC subjects); repeated stratified CV
Cohort-size scaling (Fig. S1)	up to 145 subj	up to 41 subj	balanced subsets, leave-one-patient-out
Tangent-space SVM benchmark (Fig. 3E,F)	557 sessions (143 subj)	145 sessions (40 subj)	GSR preprocessing; nested 5 × 5 subject-grouped CV (702 sessions, train-only tangent reference)

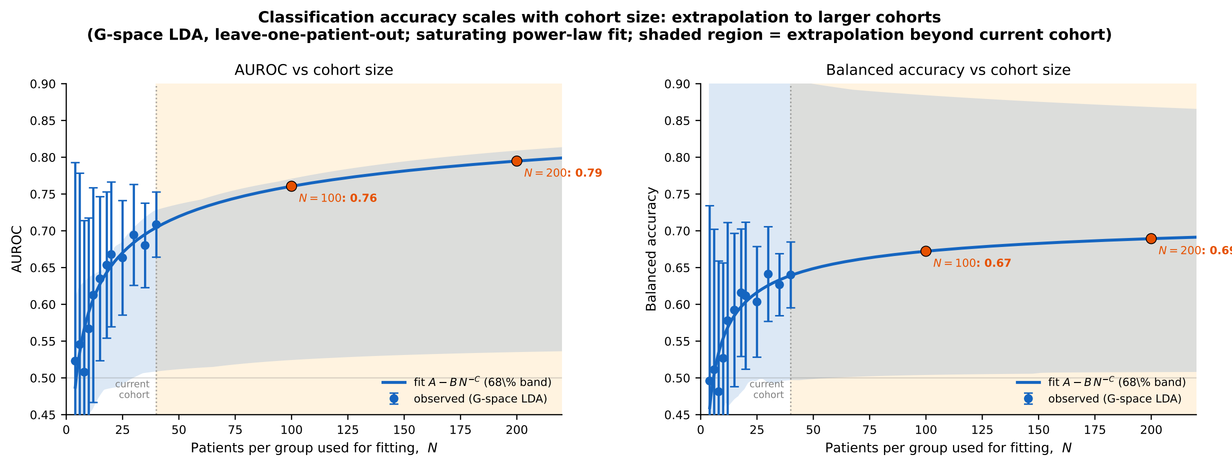

Figure S1. Classification accuracy scales with cohort size. G-space LDA balanced accuracy and AUROC (leave-one-patient-out) as a function of the number of patients per group used for fitting, N (blue points, mean ± SD over resampled subsets). A saturating power law $m(N) = A - B\,N^{-C}$ is fit to each metric (blue line; shaded 68% band from the fit covariance) and extrapolated beyond the current cohort (orange region). Accuracy has not saturated at $N = 40$: the extrapolation projects AUROC ≈ 0.76 at $N = 100$ and ≈ 0.80 at $N = 200$ (orange markers), indicating that the modest present accuracy is sample-size-limited rather than an intrinsic ceiling of the read-out representation.

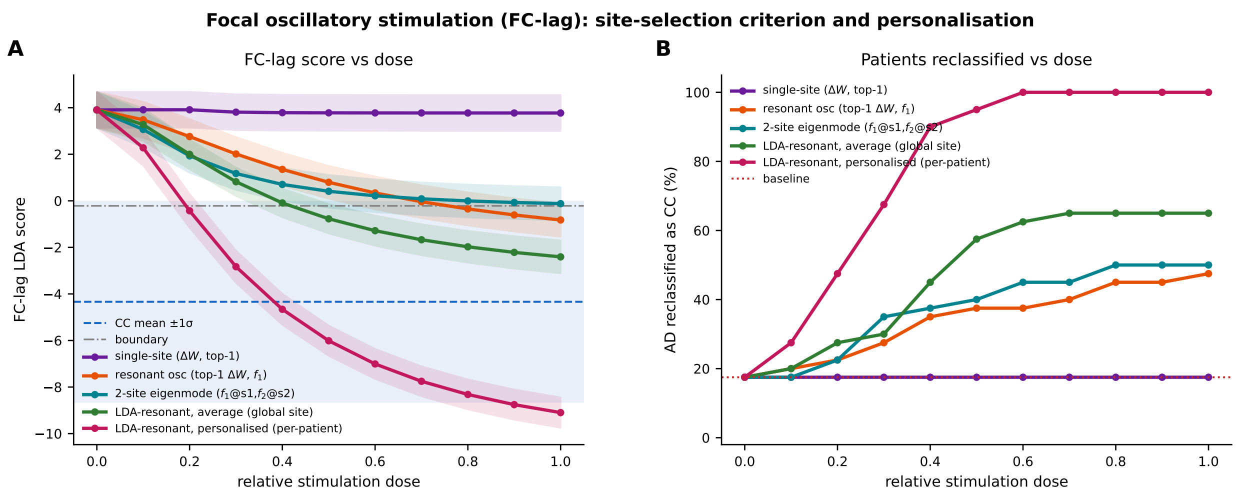

Figure S2. Focal oscillatory stimulation: site-selection criterion and personalisation (FC-lag classifier). Five focal interventions on the same seeded reservoir ($N = 40$ AD) against a common normalised dose (W-interpolation $\alpha \in [0, 5]$ for the single-site strategy; amplitude $A \in [0, 10]$ for the oscillatory drives). Strategies: single-site $\Delta\mathbf{W}$ interpolation (top-1; purple); resonant oscillatory drive at the top-1 $\Delta\mathbf{W}$ site at f_1 (orange); 2-site eigenmode-matched drive (teal; $A\sin(2\pi f_1 t)$ at the site coupling most to eigenmode 1 and $A\sin(2\pi f_2 t)$ at the site coupling most to eigenmode 2, $f_1 = 0.077$, $f_2 = 0.177$ cycles/step); and the *LDA-resonant* drive at f_1 applied at the site whose resonant stimulation moves the FC-lag discriminant most toward CC, chosen either from the population-average projection (green; single global site) or *personalised* per patient from each subject's own model (crimson). (**A**) FC-lag score and (**B**) reclassification versus dose. Blue dashed, CC mean $\pm 1\sigma$; dot-dashed, boundary. Selecting the site by its *effect on the discriminant* (LDA-resonant) is far more effective than by modal excitation (eigenmode) or disease-norm (single-site): the average LDA-resonant drive reclassifies ~65% of patients, and *personalising* the target per patient reaches ~100% (FC-lag score driven well past the CC mean), versus ~50% for the eigenmode/single-site-resonant drives and ~17% for the static single-site correction. The oscillatory drives leave $\mathbf{W}$ unchanged and so act only through the nonlinear FC-lag read-out (the linear G-space classifier, which depends on $\mathbf{W}$ alone, is by construction insensitive to them). Personalised targets are selected on each patient's own model (an upper bound on personalisation) and high amplitudes are supra-physiological; the comparison at matched dose nonetheless shows a large personalisation gain.

Figure S3. Training and test reconstruction error of the read-out archetype model. Each fitted read-out $\mathbf{W}_i$ is projected onto the top-K_{PC} right singular vectors of its reservoir states, and the projected read-outs are then compressed across patients into M SVD archetypes; the error is the signal-space normalised mean-squared error, NMSE = $\langle(\mathbf{Y} - \mathbf{X}\mathbf{W}_g^{\top})^2\rangle / \langle \mathbf{Y}^2 \rangle$, of the archetype reconstruction $\mathbf{W}_g$ ($\sigma = 0.025$; mean over 4 random 70/30 patient splits). (**A**) Training and (**B**) test NMSE over the (K_{PC}, M) grid. (**C**) Test NMSE vs M for each K_{PC}. Training error falls steeply with both K_{PC} and M, whereas test error is governed chiefly by K_{PC} (plateauing for $K_{\mathrm{PC}} \gtrsim 100$) and is nearly flat in M beyond $M \approx 10$: additional archetypes fit training-specific variance, the widening train–test gap, without improving generalisation. This justifies the projection rank $K_{\mathrm{PC}} = 200$ used throughout and shows that a low-dimensional archetype basis captures the cross-patient-shared read-out structure.

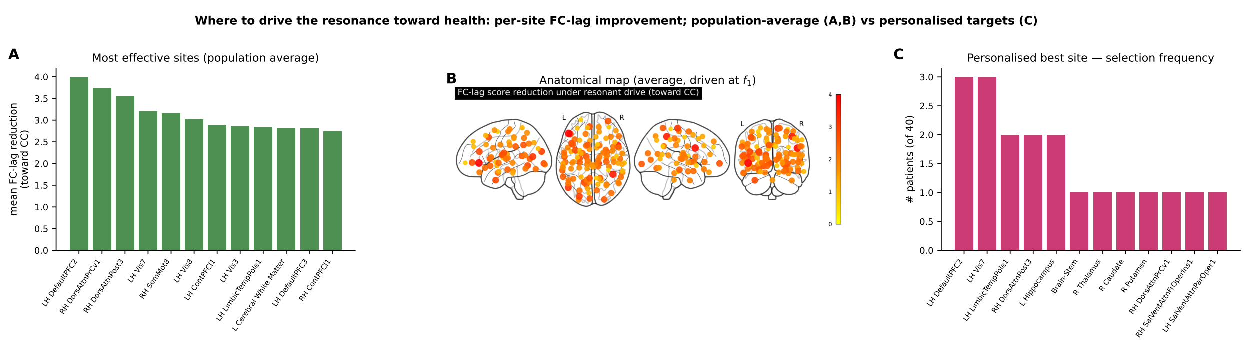

Figure S4. Where to drive the resonance toward health. Each of the 121 sites is driven at the leading eigenmode frequency f_1 (amplitude $A = 4$) and ranked by the resulting FC-lag LDA-score reduction toward CC, i.e. the projection of the resonant-stimulation effect onto the discriminant. (**A**) Population-average ranking of the most effective sites; the leading target is the left Default-network prefrontal cortex, followed by dorsal-attention, visual and somatomotor association cortex. (**B**) The same on a glass brain (marker size/colour = mean reduction): the effective sites are predominantly cortical and distributed. (**C**) When the target is chosen *per patient* (each subject's own model), the selected sites are heterogeneous, most are chosen by only a few patients and span cortical (Default, visual) and subcortical (hippocampus, brainstem, thalamus) structures, so no single global target is optimal for all, motivating personalised targeting.

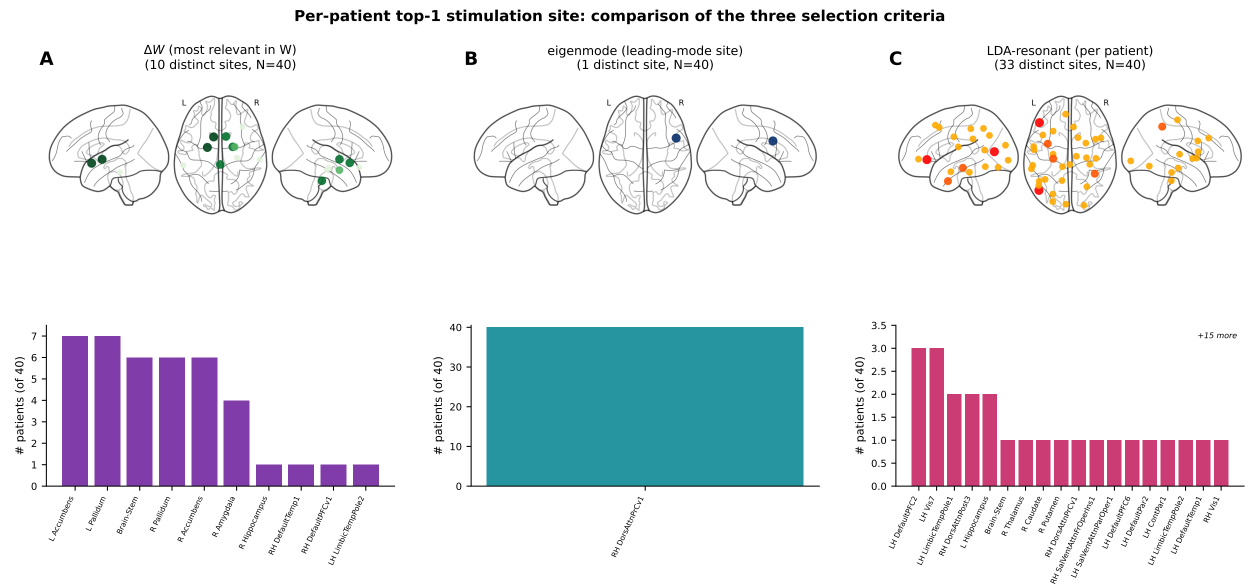

Figure S5. Per-patient top-1 stimulation site: comparison of the three selection criteria (same seeded reservoir; glass brain on top, selection-frequency histogram below; $N = 40$ AD patients). (**A**) ΔW (most relevant in the read-out; the single-site ΔW stimulation): the site of largest $\|W_{\mathrm{CC}} - W_p\|$. These targets cluster on a few *subcortical* structures shared across patients (10 distinct sites; mostly accumbens, pallidum, brain-stem, amygdala), i.e. where the disease most distorts the read-out. (**B**) *Eigenmode*: the site of maximal coupling to the leading reservoir eigenmode. Being a property of the (shared) dynamics, it is a *single global site* for all patients (right dorsal-attention cortex). (**C**) *LDA-resonant* (per patient): the site whose resonant drive moves that subject's FC-lag discriminant most toward CC. These therapeutic targets are highly heterogeneous, 33 **distinct sites for the** 40 **patients**, each selected by at most three, spanning Default-network, visual and attention cortex together with medial-temporal and subcortical structures. The three criteria answer different questions: where the disease lives in the read-out (ΔW), the fixed resonance node of the dynamics (eigenmode), and the personalised lever that drives recovery (LDA-resonant); only the last is genuinely individualised, so no single global site is adequate.

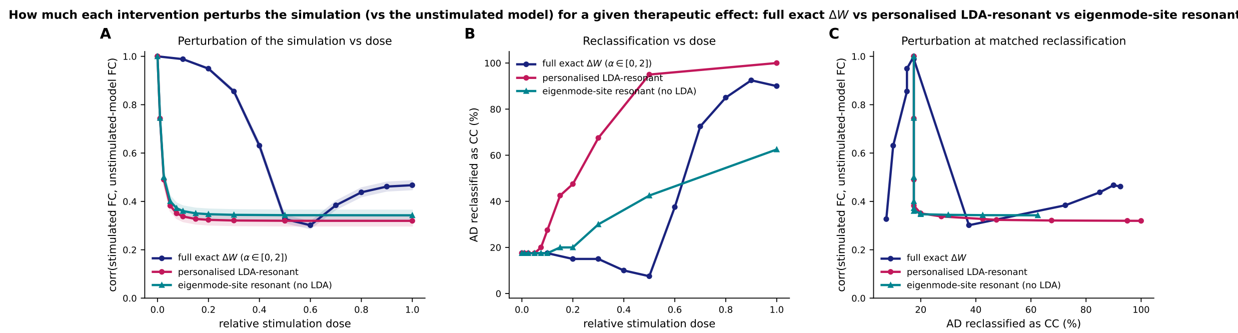

Figure S6. How much each intervention perturbs the simulation, for a given therapeutic effect: full exact ΔW, personalised LDA-resonant, and eigenmode-site resonant stimulation ($N = 40$ AD). The perturbation is the Pearson correlation (FC off-diagonals) between the *closed-loop free-run* simulated FC under stimulation (generative protocol of Fig. 2: drive 5 steps, then free-run) and the *unstimulated* model's own free-run FC, so it equals 1 at zero dose by construction and falls as the intervention reshapes the dynamics. (The unstimulated free-run reproduces the empirical FC at $r \approx 0.61$, as in Fig. 2.) Three interventions are compared: full-site ΔW interpolation $W_{\mathrm{int}} = (1-\alpha)W_p + \alpha W_{\mathrm{CC}}$ ($\alpha \in [0, 2]$; $\alpha = 1$ full replacement by the CC-mean read-out, $\alpha > 1$ over-correction), the *personalised* LDA-resonant drive (each patient's own discriminant-aligned site), and the *eigenmode-site* resonant drive (single global site of maximal coupling to the leading eigenmode, selected by modal coupling *without* the LDA); both oscillatory drives use $A \in [0, 10]$ at f_1, sampled finely at low A. (**A**) Perturbation versus normalised dose. Both single-site resonant drives fall *steeply but continuously* ($1 \to 0.74 \to 0.49 \to 0.38$ over $A = 0.1$–0.75) and saturate near $r \approx 0.33$: because the injected $A \sin$ is added to the un-scaled input whereas the data drive is $\mathit{ff} \cdot \mathrm{tgt}$ ($\mathit{ff} = 0.1$), even $A = 1$ is an order of magnitude above the normal drive, and at f_1 the closed loop amplifies it until the resonant mode's (amplitude-normalised) spatial pattern dominates the FC. This perturbation is *network-wide*, not local to the driven node: recomputing the correlation after deleting the driven site's own row and column leaves it unchanged ($r = 0.32$ vs 0.32 at $A = 10$), i.e. every parcel, not just the stimulated one, becomes phase-locked to the driven mode. (**B**) Reclassification versus dose: the full ΔW correction reverts the FC-lag classifier only with over-correction ($\alpha \gtrsim 1.4$; $\leq$8% at the exact $\alpha = 1$, since the reservoir state stays AD-driven, consistent with Fig. 4), reaching ~90% by $\alpha = 2$; the personalised LDA-resonant drive reaches 100%, whereas the eigenmode-site drive plateaus at ~62%. (**C**) Perturbation at matched reclassification. The two single-site resonant drives perturb the simulation *near-identically* (correlation to the unstimulated model ≈ 0.32–0.34 throughout), yet only the discriminant-aligned (LDA) target converts that same dynamical perturbation into full reclassification, selecting the site by its effect on the discriminant, rather than by modal coupling alone, extracts more therapeutic effect per unit of perturbation. The global ΔW overwrite perturbs less at low effect but must over-correct and rewrite all 121 read-out weights to compete.

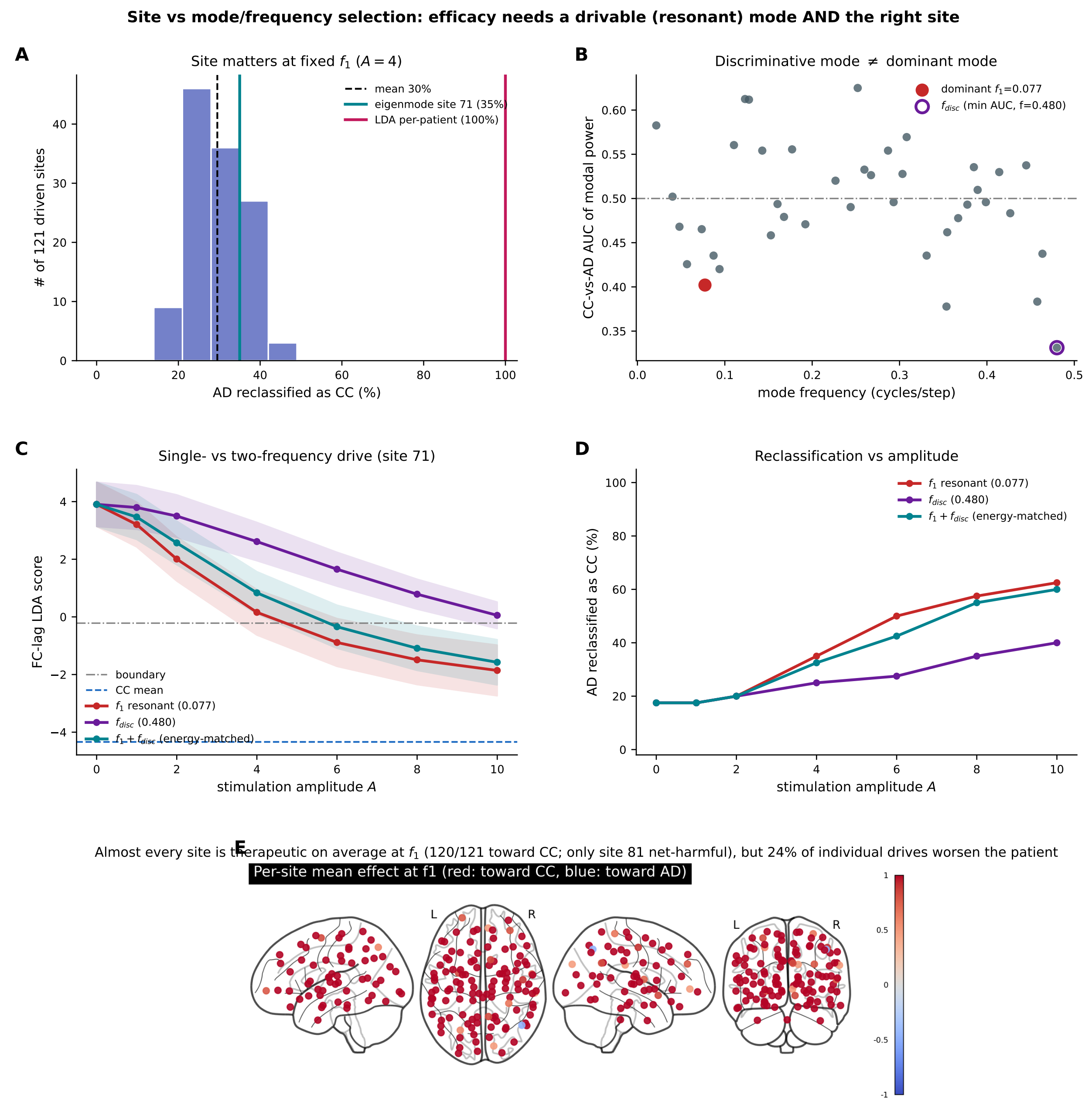

Figure S7. Why site and resonance both matter: a drivable mode is not the same as a disease-discriminative one ($N = 40$ AD). Because a resonant drive reshapes the FC network-wide (fig. S6), one might hope that the frequency/mode alone determines the effect. It does not. (**A**) Driving each of the 121 sites at the dominant-mode frequency f_1 ($A = 4$): the per-site reclassification rate spans the whole low range (15–48%, mean 30%; no site reaches ≥90%), so the *site* still decides the outcome, only per-patient (LDA) selection reaches 100%. (**B**) Ranking eigenmodes by how well their per-patient modal power separates CC from AD (AUC): the most discriminative modes lie at higher frequencies ($f \approx 0.25$–0.48), while the dominant (least-damped, most *drivable*) mode $f_1 = 0.077$ is only weakly discriminative (AUC 0.40, rank 7/40). (**C**,**D**) At a fixed site (eigenmode site 71), driving the most discriminative *deficit* mode $f_{\text{disc}} = 0.48$ (largest AD power deficit, the only correctable direction since drive can only add power) is markedly *less* effective than the resonant f_1 (40% vs 62% reclassified at $A = 10$), and an energy-matched two-frequency drive $f_1 + f_{\text{disc}}$ does *not* beat f_1 alone (60% vs 62%): diverting energy to the weakly-driven discriminative mode loses more amplification than it gains in direction. Efficacy thus requires a *drivable* (resonant) mode *and* the right site, which is precisely what the LDA-resonant criterion optimises, and why it outperforms modal-coupling, disease-deviation, or discriminative-frequency heuristics. (**E**) Signed per-site mean effect at f_1 on a glass brain (red: toward CC; blue: toward AD). Almost every site is therapeutic on average (120/121 toward CC; only the right control/parietal site is net-harmful, and only marginally). This is *not* because the drive finds the CC direction by chance: a resonant drive injects the *same* stereotyped dominant-mode pattern into every patient (site-independent; fig. S6), so there is effectively one driven direction, and it is CC-aligned because (i) AD sits at the far extreme of the discriminant (baseline score +3.9 vs boundary −0.22, CC mean −4.3) and the low-structure oscillatory FC lacks the AD-specific lag pattern, and

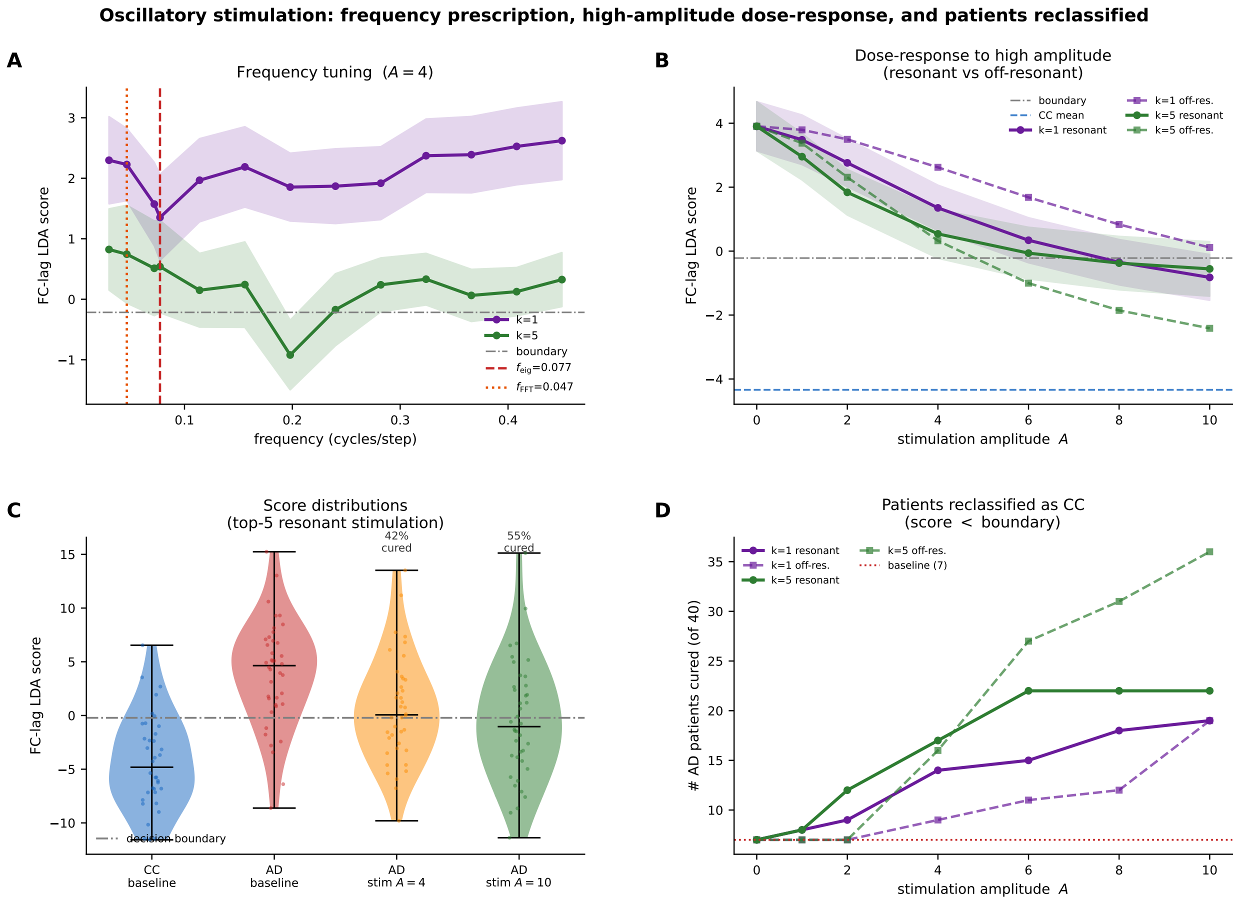

Figure S8. Frequency prescription and high-amplitude dose-response of focal oscillatory stimulation ($N = 40$ AD). A sinusoidal drive $A \sin(2\pi f t)$ is injected at the top-k most-affected (ΔW) sites and the FC-lag score re-evaluated (lower = more control-like). (**A**) Frequency tuning at moderate amplitude ($A = 4$): mean ± SEM for the single top site ($k = 1$, purple) and the top-five sites ($k = 5$, green). For $k = 1$ the score is minimal precisely at the reservoir's dominant eigenmode frequency $f_{\text{eig}} = 0.077$ (red dashed), a sharp resonance; f_{FFT} (orange dotted) marks the spectral peak of the AD reservoir states. The $k = 5$ optimum broadens and shifts, as the FC-lag effect is a nonlinear functional of the dynamics, not the modal excitation amplitude. (**B**) Dose-response to high amplitude ($A = 10$), resonant (f_{eig}, solid) vs off-resonance (highest frequency, dashed) for $k = 1$ and $k = 5$; scores move toward the CC mean (blue dashed) as amplitude grows. (**C**) FC-lag score distributions: CC baseline, AD baseline, and AD after top-five resonant stimulation at $A = 4$ and $A = 10$ (decision boundary dot-dashed; fraction crossing it annotated). (**D**) Patients reclassified as CC vs amplitude, for top-1/top-5 × resonant/off-resonance (baseline dotted): resonance is most efficient at low-to-moderate amplitude, whereas at supra-physiological amplitude even an off-resonance multi-site drive reclassifies most patients (36/40 at $A = 10$), an energy-driven (non-specific) saturation effect.

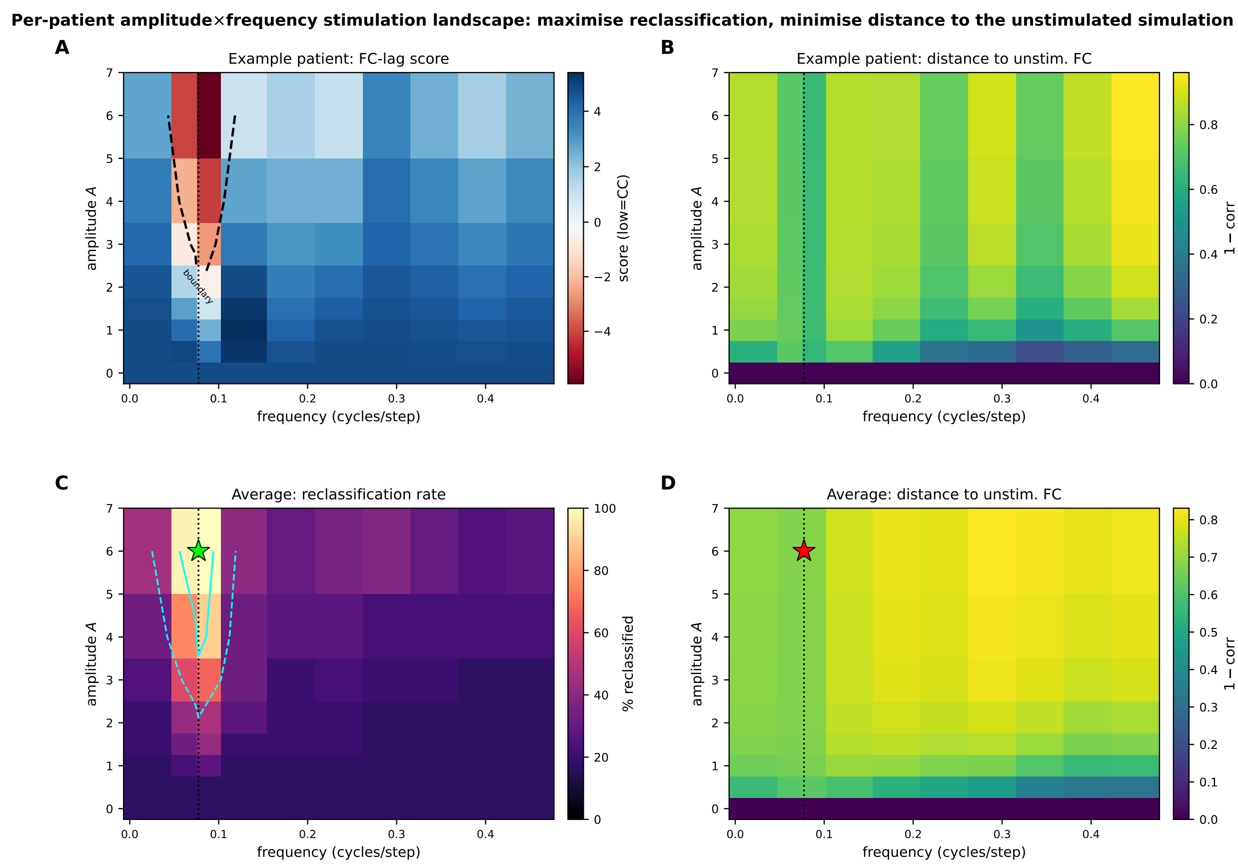

Figure S9. Per-patient amplitude×frequency stimulation landscape: trading off reclassification against departure from the unstimulated simulation ($N = 40$ AD). Each patient is driven at its personalised LDA-resonant site over an 8×10 grid of amplitude A and frequency f; we record the FC-lag score (reclassified if below the boundary; classifier space) and the distance to the *unstimulated* simulation, $1 - \text{corr}$ between the stimulated and unstimulated closed-loop free-run FC. (**A**,**B**) One representative patient (median baseline score): the FC-lag score (A; boundary contour in black) and the distance (B). The score drops below the boundary only in a narrow band of frequencies around the reservoir resonance f_1 (dotted line). (**C**) Population reclassification rate: a narrow "tongue" centred on f_1 (cyan contours at 50% and 80%), at resonance patients reclassify at the *lowest* amplitude, whereas off-resonance requires much larger drive. (**D**) Population mean distance, which grows with amplitude and is itself elevated near f_1. The two objectives therefore conflict: the operating point that maximises reclassification minus distance ($\star$; $A = 6$, $f = f_1$, 100% reclassified) sits at high distance (≈ 0.68), and reverting most patients unavoidably requires a large perturbation (distance $\gtrsim 0.66$, cf. fig. S6–S7). Resonance is nonetheless the most *efficient* axis, the minimum-distance route to any target reclassification runs along f_1, so the model's prescription is to drive the personalised site at the intrinsic resonance, accepting the minimal perturbation that crosses the boundary.